INVARIANTS OF r-SPIN TQFTS AND NON-SEMISIMPLICITY

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Abstract. For a positive integer r, an r-spin topological quantum field theory is a 2-dimensional TQFT with tangential structure given by the r-fold cover of SO_2. In particular, such a TQFT assigns a scalar invariant to every closed r-spin surface Σ. Given a sequence of scalars indexed by the set of diffeomorphism classes of all such Σ, we construct a symmetric monoidal category $\mathcal{C}$ and a $\mathcal{C}$-valued r-spin TQFT which reproduces the given sequence. We also determine when such a sequence arises from a TQFT valued in an abelian category with finite-dimensional Hom spaces. In particular, we construct TQFTs with values in super vector spaces that can distinguish all diffeomorphism classes of r-spin surfaces, and we show that the Frobenius algebras associated to such TQFTs are necessarily non-semisimple.

Keywords. Topological quantum field theory, spin structures, Frobenius algebras, symmetric monoidal categories.

Contents

1. Introduction

A basic classification result in topological quantum field theory (TQFT) states that 2-dimensional oriented *closed* TQFTs are equivalent to commutative Frobenius algebras. Similarly explicit characterisations of oriented closed TQFTs are available also in dimensions 1 and 3, but not in higher dimension. The cobordism hypothesis offers a general classification of *fully extended* TQFTs in arbitrary dimension and for arbitrary tangential structures, while there are relatively few classification results for closed TQFTs with tangential structures other than orientations. In the present paper we are concerned with one of the exceptions in dimension 2, namely *r-spin TQFTs*.

Recall that for a positive integer r, the r-spin group Spin_2^r is the r-fold cover of the rotation group SO_2, and that an r-spin surface is a surface equipped with a Spin_2^r-principal bundle together with a bundle map to the oriented orthonormal frame bundle. There is a symmetric monoidal category Bord_2^r whose morphisms are equivalence classes of r-spin bordisms, which for $r = 1$ recovers the familiar category of 2-dimensional oriented bordisms (see Section 2 for more details). An r-spin TQFT with values in a given symmetric monoidal category $\mathcal{C}$ is by definition a symmetric monoidal functor $\mathrm{Bord}_2^r \longrightarrow \mathcal{C}$. As shown in [StSz], such TQFTs are classified by "closed Λ_r-Frobenius algebras" in $\mathcal{C}$, which we review in Definition 2.7 below.

A closed r-spin surface Σ can be viewed as a morphism $\varnothing \longrightarrow \varnothing$ in Bord_2^r. Evaluating an r-spin TQFT $\mathcal{Z}\colon \mathrm{Bord}_2^r \longrightarrow \mathcal{C}$ on Σ produces the invariant $\mathcal{Z}(\Sigma) \in \mathrm{End}_{\mathcal{C}}(\mathbf{1})$. If moreover $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong \mathbb{K}$ for some field $\mathbb{K}$ of characteristic 0, then $\mathcal{Z}(\Sigma)$ gives a scalar invariant of $\mathcal{Z}$. Hence in this case, every TQFT $\mathcal{Z}\colon \mathrm{Bord}_2^r \longrightarrow \mathcal{C}$ gives rise to a sequence of scalars $(\mathcal{Z}(\Sigma))_\Sigma$, where Σ runs over all diffeomorphism classes of r-spin surfaces.

The main question that will be addressed in this paper is what kind of sequences $(\mathcal{Z}(\Sigma))_\Sigma \subset \mathbb{K}$ arise from different r-spin TQFTs. Before we give an answer in Theorem 1.2 below, we consider the more specific question of whether there are r-spin TQFTs with algebraic targets $\mathcal{C}$ that can distinguish all morphisms in Bord_2^r. In particular, it turns out that $\mathcal{C} = \mathrm{SVect}_{\mathbb{K}}$ is good enough for the second question. The choice $\mathcal{C} = \mathrm{Vect}_{\mathbb{K}}$ is not quite sufficient, but the only *super* algebra A needed is the one related to the Arf invariant (see Section 3.1). Indeed, in Section 3.7 we prove our first main result:

Theorem 1.1. For every integer r there is a non-semisimple closed Λ_r-Frobenius algebra $B^{(r)}$ (for r odd) or $C^{(r)}$ (for r even) in $\mathrm{Vect}_{\mathbb{K}}$, constructed in Sections 3.2 and 3.3. There are direct sums and tensor products of these algebras with the Arf algebra A which correspond to r-spin TQFTs valued in $\mathrm{SVect}_{\mathbb{K}}$ that have distinct invariants for all r-spin surfaces.

Put differently, we explicitly construct TQFTs in super vector spaces that can distinguish all diffeomorphism classes of r-spin surfaces. We also prove (in Section 3.8) that non-semisimplicity is a necessary condition to produce TQFTs that can distinguish all r-spin structures (if $r > 2$). This is another illustration that non-semisimple TQFTs are known or expected to resolve finer structure.[1]

The target $\mathrm{SVect}_{\mathbb{K}}$ is however not good enough to achieve the slightly more ambitious goal of constructing an r-spin TQFT $\mathcal{Z}$ such that the sequence $(\mathcal{Z}(\Sigma))_\Sigma$ recovers an arbitrary given allowed sequence of invariants. To describe a solution to this problem, and to explain what "allowed" means here, we first need to recall the classification of diffeomorphism classes of connected

[1] For example, the categorified link invariants of [KR] are computed internal to non-semisimple extended TQFTs of Landau–Ginzburg type (cf. [CMM, Thm. 3.7 & Ex. 3.13]), and semisimple TQFTs are only sensitive to stable diffeomorphism classes as shown in [RSP].

r-spin surfaces (see also Theorem 2.4). For $g \geqslant 0$, let Σ_g denote such a closed surface of genus g. Then Σ_g admits an r-spin structure if and only if $2-2g = 0 \mod r$. For $g = 0$, the sphere Σ_0 admits a unique diffeomorphism class of an r-spin structure when r is 1 or 2. For $g = 1$, there are up to diffeomorphism precisely as many r-spin structures as there are divisors d of r, and we denote the associated invariants by $\beta_d^{\mathcal{Z}}$. For $g > 1$ such that $2-2g = 0 \mod r$, Σ_g admits a single r-spin diffeomorphism class in case r is odd, and we write the corresponding invariant as $\alpha_n^{\mathcal{Z}}$, where $g = nr+1$ for some positive integer n satisfies $2-2g = 0 \mod r$. If r is even and g satisfies the condition, then up to diffeomorphism there are two r-spin structures on Σ_g, and we write $(\alpha_n^+)^{\mathcal{Z}}$ and $(\alpha_n^-)^{\mathcal{Z}}$ for the invariants that correspond to these two structures, where $g = nr/2+1$ for the appropriate integer n. (Section 2.3 has more details.)

In Section 4 we construct a family of symmetric monoidal categories, showing that if we do not impose conditions on the target category $\mathcal{C}$, then every sequence can arise from an r-spin TQFT. Our construction uses that of the symmetric monoidal categories $\mathcal{C}_\chi$ which first appeared in [Me2]. We would like to explain what happens when we restrict the target category. Using the terminology of [Me2], we call an additive $\mathbb{K}$-linear symmetric monoidal category $\mathcal{C}$ *good* if it is abelian, rigid, and has finite-dimensional Hom spaces. The second main theorem in the present paper then is the following:

Theorem 1.2. (1) Let $r \in \mathbb{Z}_{\geqslant 1}$ be odd, and let $((b_d)_{d|r}, a_1, a_2, \ldots)$ be an infinite sequence in $\mathbb{K}$. There is a good category $\mathcal{C}$ and an r-spin TQFT $\mathcal{Z}\colon \operatorname{Bord}_2^r \longrightarrow \mathcal{C}$ such that $\alpha_n^{\mathcal{Z}} = a_n$ and $\beta_d^{\mathcal{Z}} = b_d$ if and only if (for some formal variable X)

$$\sum_{n\in\mathbb{Z}_{\geqslant 1}} a_n X^n \ \in \ \operatorname{span}\left\{\frac{1}{1-\lambda X}\right\}_{\lambda\in\mathbb{K}} . \tag{1.1}$$

(2) Let $r \in \mathbb{Z}_{\geqslant 1}$ be even, and let $((b_d)_{d|r}, a_1^+, a_1^-, a_2^+, a_2^-, \ldots)$ be an infinite sequence in $\mathbb{K}$. There is a good category $\mathcal{C}$ and an r-spin TQFT $\mathcal{Z}\colon \operatorname{Bord}_2^r \longrightarrow \mathcal{C}$ such that $(\alpha_n^+)^{\mathcal{Z}} = a_n^+$, $(\alpha_n^-)^{\mathcal{Z}} = a_n^-$ and $\beta_d^{\mathcal{Z}} = b_d$ if and only if

$$\sum_{n\in\mathbb{Z}_{\geqslant 1}} a_n^+ X^n ,\ \sum_{n\in\mathbb{Z}_{\geqslant 1}} a_n^- X^n \ \in \ \operatorname{span}\left\{\frac{1}{1-\lambda X}\right\}_{\lambda\in\mathbb{K}} . \tag{1.2}$$

Notice that the theorem imposes conditions only on the values of α (corresponding to surfaces of genus $g > 1$), but not on β (corresponding to $g = 1$). The conditions (1.1) and (1.2) follow from a more general criterion on symmetric monoidal categories that are generated by an algebraic structure, constructed in [Me2] and reviewed in Sections 4.1–4.2. Here we apply that theory to the algebraic structure of closed Λ_r-Frobenius algebras.

To prove that any sequence of numbers that satisfies the conditions (1.1) or (1.2) comes from a TQFT, we will show that closed Λ_r-Frobenius algebras are closed under taking direct sums and a suitable form of tensor product. Indeed, Lemma 3.9 states that these operations commute nicely with taking invariants in the sense that for two closed Λ_r-Frobenius algebras C and D it holds that $\gamma^{C\oplus D} = \gamma^C + \gamma^D$ and $\gamma^{C\underline{\otimes} D} = \gamma^C\gamma^D$, where γ is either of the invariants α or β (where we use the equivalence of r-spin TQFTs and closed Λ_r-Frobenius algebras). This will reduce the proof of Theorem 1.2 to the construction of enough closed Λ_r-Frobenius algebras, which in turn is provided in Section 3. Of particular importance are the above-mentioned algebras B and C in $\operatorname{Vect}_{\mathbb{K}}$ as well as the Arf algebra A in $\operatorname{SVect}_{\mathbb{K}}$.

For $r = 1$ the question of what invariants one gets was studied by Khovanov, Ostrik and Kononov in [KOK]. Our results here can be considered as an extension of theirs. In Section 4.4 we explain how the categories constructed in [KOK] are related to the categories we construct here, following [Me2].

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2. r-spin surfaces, TQFTs and invariants

We summarise the notion of r-spin structures, r-spin bordisms and the category they form. Then we consider r-spin topological quantum field theories and closed Λ_r-Frobenius algebras that classify them. For a more detailed account see e.g. [StSz].

2.1. ***r*-spin surfaces.** For a positive integer r, the *r-spin group* is the r-fold covering of SO_2:

$$\begin{aligned} \mathrm{Spin}_2^r = \mathrm{SO}_2 &\longrightarrow \mathrm{SO}_2 \\ x &\longmapsto x^r \end{aligned} \tag{2.1}$$

We call a smooth 2-dimensional manifold a *surface*. An *r-spin structure on a surface* Σ is a Spin_2^r-principal bundle P over Σ together with a bundle map $q\colon P \longrightarrow F_\Sigma$ to the oriented orthonormal[2] frame bundle of Σ, which is equivariant with respect to (2.1) in the sense that $q(x.p) = x^r.q(p)$. We call such a triple (Σ, P, q) an *r-spin surface*. We note that the map $q\colon P \longrightarrow F_\Sigma$ is a $\mathbb{Z}/r$-principal bundle. Let (P, q) and (P', q') be r-spin structures over Σ. An *isomorphism of r-spin structures* $f\colon (P, q) \longrightarrow (P', q')$ is a bundle map such that the following diagram commutes:

$$\begin{tikzcd} P \arrow[rr, "f"] \arrow[dr, "q"'] & & P' \arrow[dl, "q'"] \\ & F_\Sigma & \end{tikzcd} \tag{2.2}$$

Example 2.1. Isomorphism classes of r-spin structures on a cylinder $S^1 \times \mathbb{R}$ are in bijection with $\mathbb{Z}/r$. We write S^1_x for the cylinder with r-spin structure corresponding to $x \in \mathbb{Z}/r$.

Let (Σ, P, q) and (Σ', P', q') be r-spin surfaces. A *morphism of r-spin surfaces*

$$\Phi\colon (\Sigma, P, q) \longrightarrow (\Sigma', P', q') \tag{2.3}$$

is a bundle map Φ covering a local diffeomorphism ϕ such that

$$\begin{tikzcd} P \arrow[r, "\Phi"] \arrow[d, "q"'] & P' \arrow[d, "q'"] \\ F_\Sigma \arrow[r, "(\mathrm{d}\phi)^*"] \arrow[d] & F_{\Sigma'} \arrow[d] \\ \Sigma \arrow[r, "\phi"'] & \Sigma' \end{tikzcd} \tag{2.4}$$

commutes, where the map $(\mathrm{d}\phi)^*$ is induced on frames by the differential of ϕ. If ϕ is a diffeomorphism, then we call Φ a *diffeomorphism of r-spin surfaces*; if ϕ is the identity on Σ (hence $\Sigma' = \Sigma$), then we call Φ an *isomorphism*. It follows that every isomorphism is a diffeomorphism, and that the number of diffeomorphism classes or r-spin structures on a given Σ is bounded from above by the number of isomorphism classes of Σ.

Let S_0 and S_1 be two closed smooth 1-manifolds and fix r-spin structures (P_i, q_i) on $S_i \times \mathbb{R}$ for $i \in \{0, 1\}$. Consider open neighbourhoods $U_i \subset S_i \times \mathbb{R}$ of S_i, set $U_i^+ := U_i \cap S_i \times \mathbb{R}_{\geqslant 0}$, $U_i^- := U_i \cap S_i \times \mathbb{R}_{\leqslant 0}$, and let $(P_i|_{U_i^\pm}, q_i|_{U_i^\pm})$ be the r-spin structures restricted to $U_i^\pm$. Let Σ be a

[2]Strictly speaking this requires the choice of a metric on Σ, for details we refer to e.g. [Sz, Sect. 2.2].

compact surface (with possibly nonempty boundary) with an r-spin structure (P, q). An *r-spin bordism*

$$(\Sigma, P, q)\colon (S_0 \times \mathbb{R}, P_0, q_0) \longrightarrow (S_1 \times \mathbb{R}, P_1, q_1) \tag{2.5}$$

consists of the data

$$\Big(U_0^+, P_0\big|_{U_0^+}, q_0\big|_{U_0^+}\Big) \xrightarrow{s_0^{\mathrm{in}}} (\Sigma, P, q) \xleftarrow{s_1^{\mathrm{out}}} \Big(U_1^-, P_1\big|_{U_1^-}, q_1\big|_{U_1^-}\Big)\,, \tag{2.6}$$

where the *boundary parametrisation maps* $s_i^{\mathrm{in/out}}$ are morphisms of r-spin surfaces that identify $S_0 \sqcup S_1$ with $\partial\Sigma$. We will sometimes write $\Sigma\colon S_0 \longrightarrow S_1$ for an r-spin bordism in case the r-spin structures are understood. A *diffeomorphism of r-spin bordisms* is a diffeomorphism of the compact r-spin surfaces that is compatible with the boundary parametrisation maps.

Example 2.2. The disc $D^2 \subset \mathbb{R}^2$ has a unique r-spin structure, since it is contractible. It can be an r-spin bordism in two ways:

$$D^2\colon \varnothing \longrightarrow S^1_{+1} \quad \text{or} \quad D^2\colon S^1_{-1} \longrightarrow \varnothing\,, \tag{2.7}$$

where we write $S^1_{\pm 1}$ for the codomain or domain circle $S^1 = \partial D^2$ with the r-spin structure induced from D^2.

Consider two r-spin bordisms

$$S_0 \xrightarrow{\Sigma_0} S_1 \xrightarrow{\Sigma_1} S_2 \tag{2.8}$$

where the domain of Σ_1 is equal to the codomain of Σ_0. We define the r-spin bordism

$$S_0 \xrightarrow{\Sigma_1\circ\Sigma_0} S_2 \tag{2.9}$$

by gluing the r-spin surfaces Σ_0 and Σ_1 along the boundary parametrisation maps s_1^{out} and s_1^{in}.

Definition 2.3. The *r-spin bordism category* Bord_2^r consists of the following. An object is a closed smooth 1-manifold S with an r-spin structure on $S \times \mathbb{R}$. A morphism $S_0 \longrightarrow S_1$ is a diffeomorphism class of r-spin bordisms. Composition is given by gluing r-spin bordisms, and identity morphisms are given by cylinders whose boundary parametrisations are embeddings.

The category Bord_2^r has a standard monoidal structure given by disjoint union $\sqcup$ with monoidal unit the empty set $\varnothing$, and a standard symmetric braided structure given by mapping cylinders over swap diffeomorphisms. We always view Bord_2^r as endowed with this symmetric monoidal structure.

Objects of Bord_2^r are disjoint unions of r-spin circles S^1_a, $a \in \mathbb{Z}/r$. We stress that morphisms in Bord_2^r are diffeomorphism classes, so two distinct *isomorphism* classes may correspond to the same morphism in Bord_2^r. The following theorem explains this in detail for the case of closed r-spin surfaces, which by definition represent endomorphisms $\varnothing \longrightarrow \varnothing$ in Bord_2^r. For more details see e.g. [Sz, Thm. 2.2.2] and the original references [GG, Ra].

Theorem 2.4. Let Σ_g denote a closed connected surface of genus g.

(1) There exist r-spin structures on Σ_g if and only if

$$\chi(\Sigma_g) = 2 - 2g \equiv 0 \mod r\,. \tag{2.10}$$

Assume that (2.10) holds. Then

(2) the number of isomorphism classes of r-spin structures on Σ_g is r^{2g}, and

(3) the number of diffeomorphism classes of r-spin structures on Σ_g is as follows:

g	conditions	number of diffeomorphism classes
0	$r \in \{1,2\}$	1
1	(none)	$\#\big(\text{divisors of } r\big)$
$\geqslant 2$	r even	2
	r odd	1

(2.11)

Remark 2.5. In case r is even and $g \geqslant 2$, the two diffeomorphism classes of r-spin surfaces with underlying surface Σ_g are distinguished by the Arf invariant, cf. Section 3.1 below.

2.2. **TQFTs and closed Λ_r-Frobenius algebras.** Let $\mathcal{C} \equiv (\mathcal{C}, \otimes, \mathbf{1})$ be a symmetric monoidal category, such as the category $\mathrm{Vect}_{\mathbb{K}}$ of $\mathbb{K}$-vector spaces with monoidal product $\otimes_{\mathbb{K}}$. Another example is $\mathcal{C} = \mathrm{SVect}_{\mathbb{K}}$, the category of $\mathbb{Z}/2$-graded vector spaces and even linear maps together with $\otimes_{\mathbb{K}}$, where the symmetric structure is given by $v \otimes w \longmapsto (-1)^{|v||w|} w \otimes v$, where v and w are assumed to be homogenous, and $|v|$ denotes the degree of v (and similarly for w). By a TQFT we mean a representation of a bordism category on $\mathcal{C}$. More precisely:

Definition 2.6. A *(closed) r-spin TQFT valued in* $\mathcal{C}$ is a symmetric monoidal functor $\mathcal{Z} \colon \mathrm{Bord}_2^r \longrightarrow \mathcal{C}$.

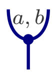
Since we are not concerned with other types of TQFTs (such as "open", "defect", or "extended") in the present paper, we drop the attribute "closed". We also reiterate that an r-spin TQFT may at best distinguish diffeomorphism classes of r-spin structures, but it cannot resolve all isomorphism classes (recall the discussion after (2.4)).

By presenting a bordism category in terms of generators and relations one can equivalently describe TQFTs in terms of algebraic structure in the target category $\mathcal{C}$. In the familiar case of oriented TQFTs, corresponding to $r = 1$, this leads to commutative Frobenius algebras. For general values of r, r-spin TQFTs give rise to the following algebraic structure:

Definition 2.7. A *closed Λ_r-Frobenius algebra C* in $\mathcal{C}$ consists of a collection of objects $C_a \in \mathcal{C}$ for $a \in \mathbb{Z}/r$ as well as morphisms

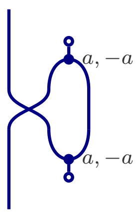
$\mu_{a,b} =$ 

$: C_a \otimes C_b \longrightarrow C_{a+b-1}\,,$ $\eta_1 =$  $: \mathbf{1} \longrightarrow C_1\,,$ (2.12)

$\Delta_{a,b} =$ 

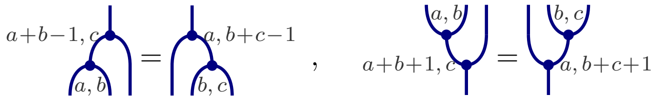
$: C_{a+b+1} \longrightarrow C_a \otimes C_b\,,$ $\varepsilon_{-1} =$  $: C_{-1} \longrightarrow \mathbf{1}$ (2.13)

for $a, b \in \mathbb{Z}/r$. The *Nakayama automorphisms* of C are

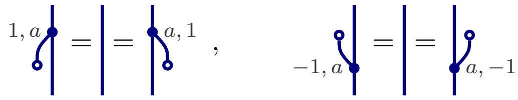
$N_a :=$ 

$: C_a \longrightarrow C_a\,.$ (2.14)

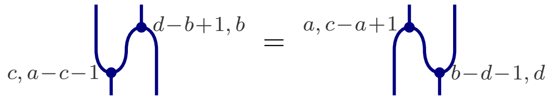
These data must satisfy the following conditions:

(co)associativity: 

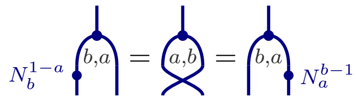
(2.15)

(co)unitality: 

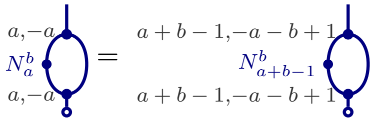
(2.16)

Frobenius relation: 

(2.17)

commutativity: 

(2.18)

twist relations: $N_a^a = 1_{C_a}$, 

(2.19)

and the deck transformation relations $N_a^r = 1_{C_a}$.

Note that a closed Λ_r-Frobenius algebra satisfies the following non-degeneracy condition: For every $a \in \mathbb{Z}/r$ there are morphisms $\delta_a := \Delta_{a,-a} \circ \eta_1 : \mathbf{1} \longrightarrow C_a \otimes C_{-a}$ such that

$$(1_{C_a} \otimes \varepsilon_{-1} \circ \mu_{-a,a}) \circ (\delta_a \otimes 1_{C_a}) = 1_{C_a} = (\varepsilon_{-1} \circ \mu_{a,-a} \otimes 1_{C_a}) \circ (1_{C_a} \otimes \delta_{-a}) \,. \tag{2.20}$$

This condition determines δ_a uniquely and provides an alternative definition of closed Λ_r-Frobenius algebras.

Assume that finite direct sums exist in $\mathcal{C}$, and consider an object $D = \bigoplus_{a\in\mathbb{Z}/r} D_a$ together with morphisms $\mu_{a,b}$, η_1 and ε_{-1} as in (2.12) and (2.13) satisfying associativity (2.15), unitality (2.16), and the non-degeneracy condition (2.20) for some morphisms δ_a. Furthermore assume that the above morphisms satisfy (2.18), (2.19) and the deck transformation relations with $\Delta_{a,-a} \circ \eta_1$ replaced by δ_a. The proof of the following lemma is completely analogous to the case of ordinary Frobenius algebras, see e.g. [Ko].

Lemma 2.8. The object D as above together with the morphisms $\mu_{a,b}$, η_1, ε_{-1} and

$$\Delta_{a,b} := (1_{D_a} \otimes \mu_{-a,a+b+1}) \circ (\delta_a \otimes 1_{D_a}) \tag{2.21}$$

form a closed Λ_r-Frobenius algebra.

In $\mathcal{C} = \mathrm{Bord}_2^r$, there is a particular closed Λ_r-Frobenius algebra, where $C_a = S_a^1$, $\mu_{a,b}$ and $\Delta_{a,b}$ are given by (upside-down) pairs-of-pants with r-structure, η_1 and ε_{-1} are given by discs as in (2.7), and N_a corresponds to a deck transformation on S_a^1. This generalises the fact that circles, pairs-of-pants and discs form a commutative Frobenius algebra in the category of 2-dimensional oriented bordisms, to which the preceding statement reduces for $r = 1$.

More generally, closed Λ_r-Frobenius algebras in an arbitrary category $\mathcal{C}$ form the objects of a category $\Lambda_r\mathrm{Frob}(\mathcal{C})$, whose morphisms $\varphi \colon C \longrightarrow D$ by definition are collections of morphisms $\varphi_a \colon C_a \longrightarrow D_a$ in $\mathcal{C}$ which preserve the structure morphisms. Analogously to the case $r = 1$, one finds that $\Lambda_r\mathrm{Frob}(\mathcal{C})$ is a groupoid. Moreover, as shown in [StSz, Cor. 5.2.2], pairs-of-pants and discs in fact generate Bord_2^r, hence closed Λ_r-Frobenius algebras "classify" r-spin TQFTs:[3]

[3]The groupoid of r-spin TQFTs valued in $\mathcal{C}$ inherits a symmetric monoidal structure from $\mathcal{C}$. Then the equivalence of [StSz] induces a (compatible) symmetric monoidal structure on $\Lambda_r\mathrm{Frob}(\mathcal{C})$ implied

Theorem 2.9. There is an equivalence of groupoids between r-spin TQFTs valued in $\mathcal{C}$ and $\Lambda_r\mathrm{Frob}(\mathcal{C})$.

To an r-spin TQFT $\mathcal{Z}$, this equivalence assigns the closed Λ_r-Frobenius algebra $C^{\mathcal{Z}}$ with $C^{\mathcal{Z}}_a := \mathcal{Z}(S^1_a)$ and whose structure maps (2.12)–(2.13) are given by the images of the generators of Bord^r_2. These generators by construction form a closed Λ_r-Frobenius algebra in Bord^r_2. Conversely, from a given object $C \in \Lambda_r\mathrm{Frob}(\mathcal{C})$ one obtains a TQFT which assigns the data of C to the generators of Bord^r_2.[4] We discuss examples of closed Λ_r-Frobenius algebras in Section 3 below.

There is a canonical construction of an r-spin TQFT from an s-spin TQFT if s divides r. Indeed, in this case we have a surjective group homomorphism $p_{r,s}\colon \mathrm{Spin}^r_2 \longrightarrow \mathrm{Spin}^s_2$ given by $x \longmapsto x^{r/s}$. Pushing forward gives us a functor

$$(p_{r,s})_*\colon \mathrm{Bord}^r_2 \longrightarrow \mathrm{Bord}^s_2 \quad (\text{for } s|r)\,, \tag{2.22}$$

hence pre-composing with $(p_{r,s})_*$ assigns an r-spin TQFT to any s-spin TQFT. On the algebraic side, this translates into a functor

$$P^*_{r,s}\colon \Lambda_s\mathrm{Frob}(\mathcal{C}) \longrightarrow \Lambda_r\mathrm{Frob}(\mathcal{C}) \quad (\text{for } s|r) \tag{2.23}$$

with

$$\big(P^*_{r,s}(C)\big)_a = C_{a \bmod s}\,, \tag{2.24}$$

and the structure maps of $P^*_{r,s}(C)$ are induced by those of $C \in \Lambda_s\mathrm{Frob}(\mathcal{C})$ in a straightforward way.

2.3. **Invariants from closed Λ_r-Frobenius algebras.** Every r-spin TQFT $\mathcal{Z}$ valued in $\mathcal{C}$ in particular provides invariants $\mathcal{Z}(\Sigma)$ of r-spin surfaces Σ. To compute them, we present Σ in terms of the generator bordisms (i.e. pairs-of-pants, discs, and mapping cylinders over deck transformations), and consider the appropriate composition of the data of the closed Λ_r-Frobenius algebra $C^{\mathcal{Z}}$ (i.e. $\mu^{\mathcal{Z}}_{a,b}, \Delta^{\mathcal{Z}}_{a,b}, \eta^{\mathcal{Z}}_1, \varepsilon^{\mathcal{Z}}_{-1}$, and $N^{\mathcal{Z}}_a$, respectively). In this section we fix a TQFT $\mathcal{Z}$ as above and compute the invariants it provides for diffeomorphism classes of closed r-spin surfaces, i.e. for morphisms $\varnothing \longrightarrow \varnothing$ in Bord^r_2.

For example, a sphere $\Sigma = S^2$ admits r-spin structures only if $r \in \{1,2\}$, of which there is then only one up diffeomorphism according to Theorem 2.4(3). Since S^2 is obtained by glueing two discs together, the associated invariant is

$$\mathcal{Z}\big(S^2\big) = \varepsilon^{\mathcal{Z}}_{-1} \circ \eta^{\mathcal{Z}}_1 =$$

$$\text{for } r \in \{1,2\}\,. \tag{2.25}$$

The most interesting case is when Σ is a torus. As described in [StSz, Sect. 4.1], every r-spin torus can be obtained by glueing two bent cylinders over an r-spin circle S^1_a after twisting with the $(1-b)$-th power of a deck transformation cylinder, for some $a, b \in \mathbb{Z}/r$. Such a torus gives rise to an endomorphism of $\varnothing \in \mathrm{Bord}^r_2$ which we denote $T(a,b)$. In line with the $g = 1$ case of Theorem 2.4(3) one finds

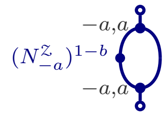
$$T(a,b) = T\big(\gcd(a,b,r), 0\big)\,. \tag{2.26}$$

in Theorem 2.9. Below in Section 3.6 we describe this tensor product of closed Λ_r-Frobenius algebras explicitly.

[4] As explained in [Sz], the case $r = 0$ corresponds to *framed* TQFTs. *Unoriented* 2-dimensional TQFTs were similarly classified in [TT], and [Cz] gives a detailed analysis of this case in analogy to the work [KOK] on the oriented case.

Hence it is enough to compute invariants of $T(d) := T(d,0)$ for $d := \gcd(a,b,r)$. For a TQFT $\mathcal{Z}$ as above these invariants are given by

$$\beta_d^{\mathcal{Z}} := \mathcal{Z}\big(T(d)\big) = \varepsilon_{-1}^{\mathcal{Z}} \circ \mu_{-a,a}^{\mathcal{Z}} \circ \big((N_{-a}^{\mathcal{Z}})^{1-b} \otimes 1_{C_a^{\mathcal{Z}}}\big) \circ \Delta_{-a,a}^{\mathcal{Z}} \circ \eta_1^{\mathcal{Z}} = \tag{2.27}$$



which does not depend on the individual values of a,b. Indeed, as shown in [StSz, Prop. 4.1.4], this is equal to the quantum dimension of $C_d^{\mathcal{Z}}$ in $\mathcal{C}$:

$$\beta_d^{\mathcal{Z}} = \dim\big(C_d^{\mathcal{Z}}\big) = \mathrm{ev}_{C_d^{\mathcal{Z}}} \circ b_{C_{-d}^{\mathcal{Z}}, C_d^{\mathcal{Z}}} \circ \mathrm{coev}_{C_d^{\mathcal{Z}}} \tag{2.28}$$

where b denotes the braiding.

According to Theorem 2.4, a closed surface of genus $g \geqslant 2$ admits an r-spin structure if and only if $2-2g \equiv 0 \mod r$. The number of diffeomorphism classes of r-spin structures then only depends on the parity of r.

We first discuss the case where r is odd. Then for fixed $g \in \mathbb{Z}_{\geqslant 2}$ a closed surface Σ_g of genus g admits precisely one diffeomorphism class of r-spin structures. To compute the invariant associated to Σ_g by $\mathcal{Z}$, we first note that the "handle map"

$$h_{x,a,b}^{\mathcal{Z}} := \mu_{a,b}^{\mathcal{Z}} \circ \Delta_{a,b}^{\mathcal{Z}} \colon C_x^{\mathcal{Z}} \longrightarrow C_{x-2}^{\mathcal{Z}} \quad \text{for } x = a+b+1 \tag{2.29}$$

may depend on the individual values of a,b, and not only on $a+b+1 \in \mathbb{Z}/r$. Choose numbers $a_i, b_i \in \mathbb{Z}/r$, $i = 0,\dots,r-1$ such that $a_i + b_i + 1 = x - 2i$. Since r is odd, the composition

$$H_x^{\mathcal{Z}} := h_{x+2,a_{r-1},b_{r-1}}^{\mathcal{Z}} \circ h_{x+4,a_{r-2},b_{r-2}}^{\mathcal{Z}} \circ \cdots \circ h_{x-2,a_1,b_1}^{\mathcal{Z}} \circ h_{x,a_0,b_0}^{\mathcal{Z}} \colon C_x^{\mathcal{Z}} \longrightarrow C_x^{\mathcal{Z}} \tag{2.30}$$

has precisely r factors. The map $H_x^{\mathcal{Z}}$ is the image under $\mathcal{Z}$ of a connected bordism $S_x^1 \longrightarrow S_x^1$ of genus r, and as shown in [Sz, Thm. 2.2.2 (3)] it only depends on $x \in \mathbb{Z}/r$, not on the individual numbers a_i and b_i. If compositions of maps of type $h_{x,a,b}$ do not depend on a,b, we suppress these indices below. Hence by composing $n \in \mathbb{Z}_{\geqslant 1}$ copies of $H_x^{\mathcal{Z}}$ and then taking the trace in $\mathcal{C}$ (where taking the trace corresponds to adding another handle by identifying the in- and outgoing boundaries S_x^1), we find that

$$\alpha_n^{\mathcal{Z}} := \mathcal{Z}(\Sigma_{nr+1}) = \mathrm{tr}\Big(\big[h_{x+2}^{\mathcal{Z}} \circ h_{x+4}^{\mathcal{Z}} \circ \cdots \circ h_{x-2}^{\mathcal{Z}} \circ h_x^{\mathcal{Z}}\big]^n\Big) \tag{2.31}$$

is the invariant associated by $\mathcal{Z}$ to Σ_{nr+1}. Note that $g := nr+1$ indeed satisfies $2-2g \equiv 0 \mod r$.

Next we assume that r is even and $g := nr/2+1 \in \mathbb{Z}_{\geqslant 2}$ for some $n \in \mathbb{Z}_{\geqslant 1}$. Then according to Theorem 2.4, Σ_g admits precisely two diffeomorphism classes of r-spin structures. These two distinct morphisms $\varnothing \longrightarrow \varnothing$ in Bord_2^r are mapped to

$$(\alpha_n^+)^{\mathcal{Z}} := \mathcal{Z}(\Sigma_{nr/2+1}^+) = \mathrm{tr}\Big(\big[(h_{x+2}^+)^{\mathcal{Z}} \circ (h_{x+4}^+)^{\mathcal{Z}} \circ \cdots \circ (h_{x-2}^+)^{\mathcal{Z}} \circ (h_x^+)^{\mathcal{Z}}\big]^n\Big) \tag{2.32}$$

$$\begin{aligned}(\alpha_n^-)^{\mathcal{Z}} := \mathcal{Z}(\Sigma_{nr/2+1}^-) = \mathrm{tr}\Big(&\big[(h_{x+2}^+)^{\mathcal{Z}} \circ (h_{x+4}^+)^{\mathcal{Z}} \circ \cdots \circ (h_{x-2}^+)^{\mathcal{Z}} \circ (h_x^+)^{\mathcal{Z}}\big]^{n-1} \\ &\circ (h_{x+2}^+)^{\mathcal{Z}} \circ (h_{x+4}^+)^{\mathcal{Z}} \circ \cdots \circ (h_{x-2}^+)^{\mathcal{Z}} \circ (h_x^-)^{\mathcal{Z}}\Big)\end{aligned} \tag{2.33}$$

where

$$(h_x^+)^{\mathcal{Z}} := \mu_{0,x-1}^{\mathcal{Z}} \circ \Delta_{0,x-1}^{\mathcal{Z}}\,, \tag{2.34}$$

$$(h_x^-)^{\mathcal{Z}} := \mu_{0,x-1}^{\mathcal{Z}} \circ \big(N_0 \otimes 1_{C_{x-1}^{\mathcal{Z}}}\big) \circ \Delta_{0,x-1}^{\mathcal{Z}}\,, \tag{2.35}$$

are both maps $C_x^{\mathcal{Z}} \longrightarrow C_{x-2}^{\mathcal{Z}}$. We can define the maps $(h_x^\pm)^{\mathcal{Z}}$ using general a,b and not necessarily $a = 0$ and $b = x-1$, as long as a is even (similarly to what we have done in the odd case). Again, $(\alpha_n^\pm)^{\mathcal{Z}}$ depends only on n, and not on the "intermediate" variables. By evaluating both morphisms on the algebra A from Section 3.1 we see that $(\alpha_n^+)^{\mathcal{Z}}$ and $(\alpha_n^-)^{\mathcal{Z}}$ correspond indeed to the two r-spin surfaces $\Sigma^{\pm}_{nr/2+1}$. By a similar argument and [Sz, Thm. 2.2.2] we have the following:

Lemma 2.10. Let r be even, and let $x \in \mathbb{Z}/r$. It holds that $h^-_{x-2}h^+_x = h^+_{x-2}h^-_x$ and $h^+_{x-2}h^+_x = h^-_{x-2}h^-_x$

In summary, an r-spin TQFT $\mathcal{Z}$ assigns the invariants (2.25), (2.28), (2.31) and (2.32) to closed r-spin surfaces, computed in terms of the associated closed Λ_r-Frobenius algebra $C^{\mathcal{Z}}$. Conversely, by dropping the superindices "$\mathcal{Z}$" in these formulas, we obtained invariants from any $C \in \Lambda_r\mathrm{Frob}(\mathcal{C})$.

Recall that for $s|r$, we have the pullback functor $P^*_{r,s}$ in (2.23). It is straightforward to compute the invariants of a TQFT associated to an image under $P^*_{r,s}$:

Lemma 2.11. Let $s|r$, $C \in \Lambda_s\mathrm{Frob}(\mathcal{C})$, and set $D := P^*_{r,s}(C)$. Then

$$\beta^D_d = \beta^C_{d \bmod s} \qquad \text{for } d|s, \tag{2.36}$$

$$\alpha^D_n = \alpha^C_{nr/s} \qquad \text{if both } r \text{ and } s \text{ are odd,} \tag{2.37}$$

$$\left(\alpha^\pm_n\right)^D = \alpha^C_{nr/(2s)} \qquad \text{if } r \text{ is even, } s \text{ is odd,} \tag{2.38}$$

$$\left(\alpha^\pm_n\right)^D = \left(\alpha^\pm_{nr/s}\right)^C \qquad \text{if both } r \text{ and } s \text{ are even.} \tag{2.39}$$

3. Examples of closed Λ_r-Frobenius algebras

In this section we present several examples of closed Λ_r-Frobenius algebras in $\mathrm{Vect}_{\mathbb{K}}$ and $\mathrm{SVect}_{\mathbb{K}}$ (Sections 3.1–3.3), which we refer to as type A, B, and C. In Section 3.6 we define the direct sum and tensor product of closed Λ_r-Frobenius algebras. This in turn is applied in Section 3.7 to algebras of type A, B, and C to construct an explicit class of examples D of closed Λ_r-Frobenius algebras whose invariants distinguish all diffeomorphism classes of r-spin surfaces. The algebras of type D are non-semisimple, and in Section 3.8 we show that, in fact, for r not a prime, every closed Λ_r-Frobenius algebra with distinct invariants is necessarily non-semisimple.

3.1. **The closed Λ_r-Frobenius algebra A related to the Arf invariant.** Let us assume that r is even, and that $\mathbb{K}$ is a field of characteristic 0. There is a closed Λ_r-Frobenius algebra $A \in \Lambda_r\mathrm{Frob}(\mathrm{SVect}_{\mathbb{K}})$ with

$$\begin{aligned}
A_x &= \mathbb{K}v_x, && \text{for } x \in \mathbb{Z}/r, \\
\eta_{+1}(1) &= v_1\,, \quad \eta_x = 0 && \text{for } x \neq 1, \\
\mu_{x,y}(v_x \otimes v_y) &= v_{x+y-1} && \text{for } x,y \in \mathbb{Z}/r, \\
\varepsilon_{-1}(v_{-1}) &= 2\,, \quad \varepsilon(v_x) = 0 && \text{for } x \neq -1 \\
\Delta_{x,y}(v_{x+y+1}) &= \frac{1}{2}v_x \otimes v_y && \text{for } x,y \in \mathbb{Z}/r, \quad \text{and} \\
N_x(v_x) &= (-1)^{1-x}v_x && \text{for } x \in \mathbb{Z}/r,
\end{aligned} \tag{3.1}$$

where the degree of v_x is $1-x \bmod 2$. The corresponding r-spin TQFT $\mathcal{Z}_A \colon \mathrm{Bord}^r_2 \longrightarrow \mathrm{SVect}_{\mathbb{K}}$ evaluated on a closed r-spin surface (Σ_g, P, q) of genus g is

$$\mathcal{Z}_A(\Sigma_g, P, q) = 2^{1-g} \cdot (-1)^{\mathrm{Arf}(\Sigma_g,P,q)}\,, \tag{3.2}$$

where $\mathrm{Arf}(\Sigma_g, P, q) \in \mathbb{Z}/2$ is the Arf invariant (see [Ra, GG], or [Sz, Sect. 3.4.2] for a review). In particular, the invariants (introduced in Section 2.3) for A are

$$\beta_d^A = \begin{cases} +1 & \text{if } d \text{ is odd} \\ -1 & \text{if } d \text{ is even} \end{cases} \quad \text{and} \quad \left(\alpha_n^{\pm}\right)^A = \pm 2^{-nr/2} . \tag{3.3}$$

3.2. **The closed Λ_r-Frobenius algebra B for $r > 1$ odd.** Let ζ be a primitive r-th root of unity. We define $B = \bigoplus_{x\in\mathbb{Z}/r} B_x \in \mathrm{Vect}_{\mathbb{K}}$ with

$$B_0 = \mathrm{span}\left(\{w_x\}_{x\in\mathbb{Z}/r} \cup \{v_0\}\right), \quad B_x = \mathrm{span}\{v_x\}_{0\neq x\in\mathbb{Z}/r} . \tag{3.4}$$

Set $u := v_1$ and $z := v_{-1}$. We define the unit by $\eta_{+1}(1) = u$, and the multiplication μ by

$$\begin{aligned} \mu(v_x \otimes v_{-x}) &= z & &\text{for } x \in \mathbb{Z}/r , \\ \mu(w_x \otimes w_{-x}) &= z \quad \text{and} \quad \mu(w_{-x} \otimes w_x) = \zeta^x z & &\text{for } 0 \leqslant x < r/2 , \\ \mu(u \otimes b) &= \mu(b \otimes u) = b & &\text{for } b \in B , \end{aligned} \tag{3.5}$$

and 0 for any other combination of basis elements. Moreover, we define the counit $\varepsilon\colon B_{-1} \longrightarrow \mathbb{K}$ by $\varepsilon_{-1}(z) = 1$, and the comultiplication Δ is defined by

$$\begin{aligned} \Delta(u) &= \sum_{x\in\mathbb{Z}/r} v_x \otimes v_{-x} + \sum_{0\leqslant x<r/2} \left(\zeta^x w_x \otimes w_{-x} + w_{-x} \otimes w_x\right) - w_0 \otimes w_0 , \\ \Delta(v_x) &= v_x \otimes z + z \otimes v_x \quad \text{for } x \neq 1 , \quad \text{and} \quad \Delta(w_x) = w_x \otimes z + z \otimes w_x . \end{aligned} \tag{3.6}$$

Lemma 3.1. *B is a closed Λ_r-Frobenius algebra in $\mathrm{Vect}_{\mathbb{K}}$.*

Proof. We verify the conditions of Lemma 2.8. Then it is straightforward to check that the comultiplication is given by (3.6).

Associativity can be checked on basis elements. Any product of three basis elements of the form $(ab)c$ or $a(bc)$ is zero unless one of a, b or c is equal to u. But if one of the basis elements is equal to u, then associativity holds because u is a left and a right unit.

The non-degeneracy condition can be checked straightforwardly for the morphisms (recall the discussion preceding Lemma 2.8)

$$\delta_a(1) := v_a \otimes v_{-a} \quad \text{for } a \neq 0 , \tag{3.7}$$

$$\delta_0(1) := v_0 \otimes v_0 + w_0 \otimes w_0 + \sum_{0<x<r/2} \left(\zeta^x w_x \otimes w_{-x} + w_{-x} \otimes w_x\right) . \tag{3.8}$$

These expressions, in turn, enable us to calculate the Nakayama automorphism (2.14) of B. For any $x \in \mathbb{Z}/r$ we get

$$\begin{aligned} N_x(v_x) &= (1 \otimes \varepsilon_{-1} \circ \mu_{x,-x})(\sigma_{B_x,B_x} \otimes 1)(v_x \otimes \delta_x(1)) = v_x \varepsilon(v_x v_{-x}) = v_x , \\ N_0(w_x) &= (1 \otimes \varepsilon_{-1} \circ \mu_{0,0})(\sigma_{B_x,B_x} \otimes 1)(w_x \otimes \delta_0(1)) \\ &= \sum_{0\leqslant y<r/2} \left(\zeta^y w_y \varepsilon(w_x w_{-y}) + w_{-y} \varepsilon(w_x w_y)\right) . \end{aligned} \tag{3.9}$$

Hence for both $0 \leqslant x < r/2$ and $r/2 < x < r$ we find $N_0(w_x) = \zeta^x w_x$.

The commutativity axiom can be verified directly, using the fact that most of the products of basis vectors are zero. The only non-trivial part of the calculation is that for $0 \leqslant x < r/2$ we have $w_{-x} w_x = \zeta^x z = N_0^{-1}(w_x) w_{-x} = w_x N_0(w_{-x})$.

For the twist axiom we first compute $f(x,k) := \mu_{x,-x}(N_x^k \otimes 1)\delta_x(1)$. For $x \neq 0$ we get $f(x,k) = \mu_{x,-x}(N_x^k \otimes 1)(v_x \otimes v_{-x}) = z$, while for $x = 0$ we get

$$\begin{aligned} f(0,k) &= \mu_{0,0}(N_0^k)\Big(v_0 \otimes v_0 + \sum_{0 \leqslant x < r/2} (\zeta^x w_x \otimes w_{-x} + w_{-x} \otimes w_x) - w_0 \otimes w_0\Big) \\ &= z + \sum_{0 \leqslant x < r/2} \big(\zeta^{x+kx} z + \zeta^{-x-kx} z - z\big) \\ &= z\Big(\sum_{x \in \mathbb{Z}/r} \zeta^{(1+k)x} + 1\Big) = \begin{cases} z & \text{if } k \neq -1 \\ (r+1)z & \text{if } k = -1 \end{cases} . \end{aligned} \tag{3.10}$$

We need to show that $f(x,k) = f(x-k-1,k)$ for every $x,k \in \mathbb{Z}/r$. If $k \neq -1$, then both $f(x,k) = z$ and $f(x-k-1,k) = z$. If $k = -1$, then $f(x-(-1)-1,-1) = f(x,-1)$, and we are done. □

Finally we compute invariants (recall Section 2.3).

Lemma 3.2. The invariants of the closed Λ_r-Frobenius algebra B are

$$\beta_d^B = \begin{cases} 1 & \text{if } d \neq r, \\ r+1 & \text{if } d = r, \end{cases} \quad \text{and} \quad \alpha_n^B = 0\,.$$

Proof. From Equation (2.28) we know that $\beta_d = \dim(B_d)$, and the first part of the lemma already follows from the definition of B. For the second part, recall the operators h_x and H_x in (2.29) and (2.30), respectively, and the definition $\alpha_n = \operatorname{tr}(H_x^n)$ from (2.31). However, if $x \neq 1$, then $m_{a,b}\Delta_{a,b}(v_x) = 0$, and in particular $h_x = 0$ and $H_x = 0$. □

Notation 3.3. To stress the value of r, we denote the algebra constructed above by $B^{(r)}$.

3.3. **The closed Λ_r-Frobenius algebra C for $r > 2$ even.** Let ζ again be a primitive r-th root of unity. We define $C = \bigoplus_{x \in \mathbb{Z}/r} C_x \in \mathrm{Vect}_{\mathbb{K}}$ with

$$C_0 = \operatorname{span}\big(\{w_{i,j}\}_{i \in \{0,1\}, j \in \mathbb{Z}/r} \cup \{v_0\}\big)\,, \quad C_x = \operatorname{span}\{v_x\}_{0 \neq x \in \mathbb{Z}/r}\,. \tag{3.11}$$

Write $u := v_1$ and $z := v_{-1}$. We define the unit by $\eta_{+1}(1) = u$, and the multiplication μ by

$$\begin{aligned} \mu(v_x \otimes v_{-x}) &= z && \text{for } x \in \mathbb{Z}/r\,, \\ \mu(w_{0,x} \otimes w_{1,-x}) = z \quad &\text{and} \quad \mu(w_{1,-x} \otimes w_{0,x}) = \zeta^{-x} z && \text{for } x \in \mathbb{Z}/r\,, \\ \mu(u \otimes c) &= \mu(c \otimes u) = c && \text{for } c \in C\,, \end{aligned} \tag{3.12}$$

and 0 for any other combination of basis elements. We define the counit by $\varepsilon_{-1}(z) = 1$, and the comultiplication Δ acts non-trivially as follows:

$$\begin{aligned} \Delta(u) &= \sum_{x \in \mathbb{Z}/r} \Big(v_x \otimes v_{-x} + \zeta^x w_{0,x} \otimes w_{1,-x} + w_{1,-x} \otimes w_{0,x}\Big)\,, \\ \Delta(v_x) &= v_x \otimes z + z \otimes v_x \quad \text{for } x \neq 1\,, \\ \Delta(w_{0,x}) &= w_{0,x} \otimes z + z \otimes w_{0,x} \quad \text{and} \quad \Delta(w_{1,x}) = w_{1,-x} \otimes z + z \otimes w_{1,-x}\,. \end{aligned} \tag{3.13}$$

Lemma 3.4. C is a closed Λ_r-Frobenius algebra in $\mathrm{Vect}_{\mathbb{K}}$.

Proof. Associativity and unitality can be checked analogously to the case of B in Section 3.2. The non-degeneracy condition can be shown for the morphisms

$$\delta_a(1) := v_a \otimes v_{-a} \quad \text{for } a \neq 0 \text{ and} \tag{3.14}$$

$$\delta_0(1) := v_0 \otimes v_0 + \sum_{x \in \mathbb{Z}/r} \Big(\zeta^x w_{0,x} \otimes w_{1,-x} + w_{1,-x} \otimes w_{0,x}\Big)\,, \tag{3.15}$$

which can be checked to correspond to Δ in (3.13).

We calculate the Nakayama automorphism of C analogously to the case of B. For $x \in \mathbb{Z}/r$ we find

$$N_x(v_x) = v_x\,, \quad N_0(w_{0,x}) = \zeta^x w_{0,x}\,, \quad N_0(w_{1,x}) = \zeta^x w_{1,x}\,. \tag{3.16}$$

For the commutativity axiom, we only need to take care of the multiplications that are non-zero. We use the formulas for the Nakayama automorphism and get

$$\begin{aligned} v_x v_{-x} &= N_x^{x-1}(v_{-x})v_x = v_{-x}N_{-x}^{1-x}(v_x) = z\,,\\ w_{0,x}w_{1,-x} &= N_0^{-1}(w_{1,-x})w_{0,x} = w_{1,-x}N_0(w_{0,x}) = z\,,\\ w_{1,-x}w_{0,x} &= N_0^{-1}(w_{0,x})w_{1,-x} = w_{0,x}N_0(w_{1,-x}) = \zeta^{-x}z\,. \end{aligned} \tag{3.17}$$

For the twist axiom, we write as before $f(x,k) := m_{x,-x}(N_x^k \otimes 1)\Delta_{x,-x}(u)$. For $x \neq 0$ we have $f(x,k) = \mu_{x,-x}(N_x^k \otimes 1)(v_x \otimes v_{-x}) = z$. For $x = 0$ we have

$$\begin{aligned} f(0,k) &= \mu_{0,0}(N_0^k \otimes 1)\Big(v_0 \otimes v_0 + \sum_{x\in\mathbb{Z}/r}(w_{1,-x}\otimes w_{0,x} + \zeta^x w_{0,x}\otimes w_{1,-x})\Big)\\ &= z + \sum_{x\in\mathbb{Z}/r}\zeta^{-x-kx}z + \zeta^{x+kx}z = \Big(2\sum_{x\in\mathbb{Z}/r}\zeta^{(1+k)x} + 1\Big)z\\ &= \begin{cases} z & \text{if } k \neq -1,\\ (2r+1)z & \text{if } k = -1. \end{cases} \end{aligned} \tag{3.18}$$

We need to show that $f(x,k) = f(x-k-1,k)$ for every $x,k \in \mathbb{Z}/r$. If $k=-1$, then this equality holds trivially, and if $k \neq -1$, then $f(x,-1) = f(x-(-1)-1,-1)$. □

The proof of Lemma 3.2 generalises directly to a proof of the following:

Lemma 3.5. The invariants of the closed Λ_r-Frobenius algebra C are

$$\beta_d^C = \begin{cases} 1 & \text{if } d \neq r,\\ 2r+1 & \text{if } d = r, \end{cases} \quad \text{and} \quad \left(\alpha_n^{\pm}\right)^C = 0\,. \tag{3.19}$$

Notation 3.6. To stress the value of r, we denote the algebra constructed above by $C^{(r)}$.

3.4. **A one-parameter family of 1-dimensional algebras.** Let $\kappa \in \mathbb{K}^\times$, and consider the 1-dimensional algebra $\mathbb{K}$. We define a counit by $\varepsilon(1) = \kappa^{-1}$. This gives us a commutative Frobenius algebra, where $\Delta(1) = \kappa 1 \otimes 1$. The handle endomorphism is then multiplication by κ. Pulling back with $P^*_{r,1}$ from (2.23), we get a Λ_r-Frobenius algebra which we denote by E_κ.

Lemma 3.7. If r is odd, then the invariants of E_κ are

$$\beta_d^{E_\kappa} = 1\,, \quad \alpha_n^{E_\kappa} = \kappa^{rn} \qquad \text{for all } d|r \text{ and } n \in \mathbb{Z}_{\geqslant 1}. \tag{3.20}$$

If r is even, then the invariants of E_κ are

$$\beta_d^{E_\kappa} = 1\,, \quad (\alpha_n^+)^{E_\kappa} = (\alpha_n^-)^{E_\kappa} = \kappa^{rn/2} \qquad \text{for all } d|r \text{ and } n \in \mathbb{Z}_{\geqslant 1}. \tag{3.21}$$

Proof. Using Lemma 2.11, we may forget about the r-spin structure, and use the fact that the handle endomorphism H is multiplication by κ, and therefore the trace of H^{rn} is κ^{rn}. □

3.5. **Another non-semisimple algebra.** Consider the algebra $\mathbb{K}[x]/(x^2)$. This is a commutative Frobenius algebra with $\varepsilon(1) = 0$ and $\varepsilon(x) = 1$. We calculate the comultiplication as before to get $\Delta(1) = 1 \otimes x + x \otimes 1$ and $\Delta(x) = x \otimes x$. The handle endomorphism is then given by $1 \longmapsto 2x$, $x \longmapsto 0$, and is therefore nilpotent. We denote the pullback of this algebra to a closed Λ_r-Frobenius algebra by F.

Lemma 3.8. If r is odd, then the invariants of F are

$$\beta_d^F = 2\,, \quad \alpha_n^F = 0 \qquad \text{for all } d|r \text{ and } n \in \mathbb{Z}_{\geqslant 1}. \tag{3.22}$$

If r is even, then the invariants of F are

$$\beta_d^F = 2\,, \quad (\alpha_n^+)^F = (\alpha_n^-)^F = 0 \qquad \text{for all } d|r \text{ and } n \in \mathbb{Z}_{\geqslant 1}. \tag{3.23}$$

Proof. This follows exactly by the same reasoning as the proof of Lemma 3.7, using the fact that the handle endomorphism is nilpotent. □

3.6. **Operations on algebras.** Let $X = \bigoplus_{x\in\mathbb{Z}/r} X_x$ and $Y = \bigoplus_{x\in\mathbb{Z}/r} Y_x$ be two closed Λ_r-Frobenius algebras in $\mathrm{Vect}_{\mathbb{K}}$, with respective structure maps $(\mu^X_{x,y}, \Delta^X_{x,y}, \eta^X_1, \varepsilon^X_{-1})$ and $(\mu^Y_{x,y}, \Delta^Y_{x,y}, \eta^Y_1, \varepsilon^Y_{-1})$. We set

$$X \oplus Y := \bigoplus_{x\in\mathbb{Z}/r} (X_x \oplus Y_x)\,, \qquad X \underline{\otimes} Y := \bigoplus_{x\in\mathbb{Z}/r} (X_x \otimes_{\mathbb{K}} Y_x)\,. \tag{3.24}$$

Notice that in general $X\underline{\otimes}Y$ is a proper subspace of $X \otimes Y$. Both $X \oplus Y$ and $X\underline{\otimes}Y$ naturally inherit the structure of a closed Λ_r-Frobenius algebra from X and Y:

Lemma 3.9. (i) $(X \oplus Y, \mu^X_{x,y} \oplus \mu^Y_{x,y}, \Delta^X_{x,y} \oplus \Delta^Y_{x,y}, \eta^X_1 \oplus \eta^Y_1, \varepsilon^X_{-1} \oplus \varepsilon^Y_{-1}) \in \Lambda_r\mathrm{Frob}(\mathrm{Vect}_{\mathbb{K}})$.
(ii) $(X\underline{\otimes}Y, \mu^X_{x,y} \otimes \mu^Y_{x,y}, \Delta^X_{x,y} \otimes \Delta^Y_{x,y}, \eta^X_1 \otimes \eta^Y_1, \varepsilon^X_{-1} \otimes \varepsilon^Y_{-1}) \in \Lambda_r\mathrm{Frob}(\mathrm{Vect}_{\mathbb{K}})$.
(iii) If Σ is a closed connected r-spin surface, then $\mathcal{Z}_{X\oplus Y}(\Sigma) = \mathcal{Z}_X(\Sigma) + \mathcal{Z}_Y(\Sigma)$ and $\mathcal{Z}_{X\underline{\otimes}Y}(\Sigma) = \mathcal{Z}_X(\Sigma)\mathcal{Z}_Y(\Sigma)$.

Proof. The fact that $X \oplus Y$ and $X\underline{\otimes}Y$ both satisfy the axioms of a closed Λ_r-Frobenius algebra is a direct verification, and is based on the fact that all the axioms are given by equalities between connected diagrams.

For part (iii) about the invariants, we first notice that $N^{X\oplus Y}_x \colon X_x \oplus Y_x \longrightarrow X_x \oplus Y_x$ is given by $N^X_x \oplus N^Y_x$. This is a straightforward verification that follows from the definition of the Nakayama automorphism. Similarly, $N^{X\underline{\otimes}Y}_x \colon X_x \otimes Y_x \longrightarrow X_x \otimes Y_x$ is equal to $N^X_x \otimes N^Y_x$. If Σ is a surface of genus 1, then there is a divisor d of r such that the invariant associated to Σ by any given closed Λ_r-Frobenius algebra is $\varepsilon\mu_{0,0}(N^{d-1} \otimes 1)\Delta_{0,0}(u)$. Using the definition of structure maps we get

$$\begin{aligned}
\mathcal{Z}_{X\oplus Y}(\Sigma) &= \varepsilon^{X\oplus Y}\mu^{X\oplus Y}_{0,0}\big((N^{X\oplus Y})^{d-1} \otimes 1\big)\Delta^{X\oplus Y}_{0,0}u^{X\oplus Y} \\
&= \varepsilon^X\mu^X_{0,0}\big((N^X)^{d-1} \otimes 1\big)\Delta^X_{0,0}u^Y + \varepsilon^Y\mu^Y_{0,0}\big((N^Y)^{d-1} \otimes 1\big)\Delta^Y_{0,0}u^Y \\
&= \mathcal{Z}_X(\Sigma) + \mathcal{Z}_Y(\Sigma)\,.
\end{aligned} \tag{3.25}$$

A similar calculation shows that $\mathcal{Z}_{X\underline{\otimes}Y}(\Sigma) = \mathcal{Z}_X(\Sigma)\mathcal{Z}_Y(\Sigma)$. If the genus of Σ is larger than 1, we use the fact that $\mathcal{Z}(\Sigma) = \mathrm{tr}(H^n_x)$ for some $n > 0$ if r is odd, and of the form $\mathrm{tr}((H^+_x)^n)$ or $\mathrm{tr}((H^+_x)^{n-1}H^-_x)$ for some $n > 0$ if r is even (recall (2.30)–(2.33)). One computes that $H^{X\oplus Y}_x = H^X_x \oplus H^Y_x$ and $H^{X\underline{\otimes}Y}_x = H^X_x \otimes H^Y_x$ for r odd, and similar identities hold when r is even. □

3.7. **A closed Λ_r-Frobenius algebra with distinct torus invariants.** First let us assume that $r > 1$ is odd and has prime factorisation

$$r = p_1^{l_1} \cdots p_k^{l_k} \tag{3.26}$$

with increasingly ordered primes $2 < p_1 < \cdots < p_k$ and exponents $l_1, \ldots, l_k > 0$. Consider the closed Λ_r-Frobenius algebra

$$D := \bigoplus_{i=1}^{k} (D^{(p_i)})^{\oplus c_i}\,, \quad \text{where} \quad D^{(p_i)} = \bigoplus_{m=1}^{l_i} P^*_{r,p_i^m}(B^{(p_i^m)})\,, \tag{3.27}$$

$$c_1 = 1\,, \tag{3.28}$$

$$c_{i+1} = c_i \cdot (\beta_r^{D^{(p_i)}} + 1), \tag{3.29}$$

the algebras $B^{(p_i^m)}$ are as defined in Section 3.2 and the torus invariants β_d for $d|r$ are defined in Equation (2.27).

Proposition 3.10. The torus invariants of D are pairwise distinct.

Proof. We start with computing the invariants of $D^{(p_i)}$. First, by Lemma 2.11 we have

$$\beta_d^{P^*_{r,p_i^m}(B^{(p_i^m)})} = \begin{cases} 1 & \text{if } p_i^m \nmid d, \\ 1 + p_i^m & \text{if } p_i^m | d. \end{cases} \tag{3.30}$$

Applying Lemma 3.9(iii) to Lemma 3.2, this gives us

$$\beta_d^{D^{(p_i)}} = \sum_{\substack{1\leqslant m \leqslant l_i \\ p_i^m \nmid d}} 1 + \sum_{\substack{1\leqslant m \leqslant l_i \\ p_i^m | d}} (1 + p_i^m) = l_i + \sum_{\substack{1\leqslant m \leqslant l_i \\ p_i^m | d}} p_i^m\,, \tag{3.31}$$

and hence

$$\beta_d^D = \sum_{1\leqslant i \leqslant k} c_i \beta_d^{D^{(p_i)}} = \sum_{1\leqslant i \leqslant k} c_i \Big(l_i + \sum_{\substack{1\leqslant m \leqslant l_i \\ p_i^m | d}} p_i^m \Big)\,. \tag{3.32}$$

We need to show that if d and e are distinct divisors of r, then $\beta_d^D \neq \beta_e^D$. Write $d = p_1^{a_1} \cdots p_k^{a_k}$ and $e = p_1^{b_1} \cdots p_k^{b_k}$. Let i be the minimal index such that $a_i \neq b_i$. Without loss of generality, we assume that $a_i < b_i$. It holds that $\beta_d^{D^{(p_j)}} = \beta_e^{D^{(p_j)}}$ for $j < i$. From (3.32) we get

$$\beta_d^D - \beta_e^D = c_i \big(\beta_d^{D^{(p_i)}} - \beta_e^{D^{(p_i)}} \big) + \sum_{j=i+1}^{k} c_j \big(\beta_d^{D^{(p_j)}} - \beta_e^{D^{(p_j)}} \big). \tag{3.33}$$

By definition of the numbers c_i it holds that $c_i | c_j$ for $i < j$. Reducing the above equation modulo c_{i+1} gives

$$\beta_d^D - \beta_e^D = c_i \big(\beta_d^{D^{(p_i)}} - \beta_e^{D^{(p_i)}} \big) \bmod c_{i+1}. \tag{3.34}$$

Dividing by c_i and using again the definition of c_{i+1} and (3.31), we get

$$\frac{1}{c_i} \big(\beta_d^D - \beta_e^D \big) = \sum_{a_i < k \leqslant b_i} p_i^k \mod \beta_r^{D^{(p_i)}} + 1\,. \tag{3.35}$$

Since the number $\sum_{a_i<k\leqslant b_i} p_i^k$ is a positive integer that is less than $\beta_r^{D^{(p_i)}} + 1$, it holds that the difference $\beta_d^D - \beta_e^D$ is non-zero as well, and we are done. □

In case r is even, one just uses the algebras C instead of B. The proof that the resulting algebra has distinct torus invariants is similar. Together with Theorem 2.4 and the discussion in Section 2.3, this proves Theorem 1.1.

3.8. **Semisimplicity and torus invariants.** The algebras $B^{(r)}$ and $C^{(r)}$ that are used to provide distinct invariants of r-spin tori are both non-semisimple. In this section we show that for a semisimple algebra the torus invariants can have at most two distinct values. We assume that the field $\mathbb{K}$ is algebraically closed.

We call a closed Λ_r-Frobenius algebra $D = \bigoplus_{x\in\mathbb{Z}/r} D_x$ *semisimple* if D as an algebra with product $\mu = \sum_{x,y\in\mathbb{Z}/r} \mu_{x,y}$ and unit $\eta = \eta_{+1}$ is semisimple. Consider a closed Λ_r-Frobenius algebra D and define a new grading by $D'_a := D_{1-a}$. By setting $\mu'_{a,b} := \mu_{1-a,1-b}\colon D_{1-a} \otimes D_{1-b} = D'_a \otimes D'_b \longrightarrow D'_{a+b} = D_{1-a-b} = D_{1-a+1-b-1}$ and $\eta' := \eta_{+1}\colon \mathbb{K} \longrightarrow D_{+1} = D'_0$ we obtain a $\mathbb{Z}/r$-associative graded algebra. The counit on the original algebra D induces a counit $\varepsilon' := \varepsilon_{-1}\colon D_{-1} = D'_0 \longrightarrow \mathbb{K}$ of degree -2.

Theorem 3.11. Let D be a semisimple closed Λ_r-Frobenius algebra in $\mathrm{Vect}_\mathbb{K}$ or in $\mathrm{SVect}_\mathbb{K}$. Then for all $a \in \mathbb{Z}/r$ we have

$$D_a \cong \begin{cases} D_1 & \text{if } r \text{ is odd,} \\ D_1 & \text{if } r \text{ is even and } a \text{ is odd,} \\ D_0 & \text{if } r \text{ is even and } a \text{ is even.} \end{cases} \tag{3.36}$$

Proof. In both cases D is a direct sum of matrix algebras. By [Ko, Lem. 2.2.8] the counit ε' on D' is of the form $\varepsilon = \mathrm{tr}(X \cdot -)$, where $X \in D'$ is invertible, and tr is the trace on the regular representation. Since ε' is of degree -2, X is of degree $+2$, i.e. we have isomorphisms $\mu(X \otimes -)\colon D'_a \longrightarrow D'_{a+2}$ for every $a \in \mathbb{Z}/r$.

In particular we have $D'_a \cong D'_0$ for every $a \in \mathbb{Z}/r$ if r is odd. If r is even, we have $D'_a \cong D'_0$ if a is even and $D'_a \cong D'_1$ if a is odd. □

Corollary 3.12. The torus invariants of a semisimple closed Λ_r-Frobenius algebra in $\mathrm{Vect}_\mathbb{K}$ or in $\mathrm{SVect}_\mathbb{K}$ can take at most two distinct values.

Proof. This follows immediately from Equation (2.28). □

4. Symmetric monoidal categories, invariants, and TQFTs

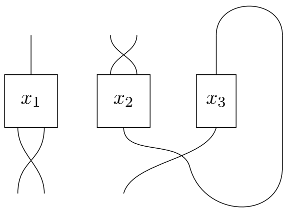
In this section we answer the question about r-spin invariants stated in the introduction, and prove Theorem 1.2 by using symmetric monoidal categories that are freely generated by an algebraic structure.

4.1. **Basic notions.** We start by recalling the relevant definitions and results from [Me2]. Let $\mathcal{C}$ be a symmetric monoidal category which we further assume to be $\mathbb{K}$-linear and rigid (hence in particular all Hom sets are $\mathbb{K}$-vector spaces, and every object has a chosen dual). An *algebraic structure* in $\mathcal{C}$ consists of an object $W \in \mathcal{C}$ together with a choice of morphisms, called *structure tensors*, $x_i\colon W^{\otimes q_i} \longrightarrow W^{\otimes p_i}$ for non-negative integers q_i, p_i. We call the tuple $((p_i, q_i))$ the *type* of $(W, (x_i))$. If (x_i) are clear from the context, we write W instead of $(W, (x_i))$.

For example, algebraic structures of type $((1, 1)^k)$ are just representations of the free algebra on k generators. Algebraic structures of type $((1, 2))$ include Lie algebras and associative algebras. Algebraic structures of type $((1, 2), (2, 1), (0, 1), (1, 0))$ include bialgebras and Frobenius algebras (with structure maps $\Delta, \mu, \eta, \varepsilon$, respectively). A Hopf algebra is an algebraic structure of type $((1, 2), (2, 1), (0, 1), (1, 0), (1, 1))$, where the extra structure tensor corresponds to the antipode.

A closed Λ_r-Frobenius algebra is an algebraic structure with underlying object W and structure tensors $(P_0, \dots, P_{r-1}, \mu, \Delta, \varepsilon, \eta)$ where, for $i \in \{0, \dots, r-1\}$, $P_i\colon W \longrightarrow W$ is an idempotent and $W = \bigoplus_i \mathrm{Im}(P_i)$. We write $W_i = \mathrm{Im}(P_i)$. The structure tensor μ then stands for $\sum_{i,j,k} \mu_{i,j}^k$, and Δ is interpreted similarly. To avoid cumbersome notation we often just write the collection of our structure tensors as (x_i) and denote the type of our algebraic structure by $((p_i, q_i))$.

A *diagram* contains boxes and strings. Every box in the diagram is labelled by one of the structure tensors or 1_W. A box that is labelled by 1_W has one input string and one output string, and a box labelled by the structure tensor x_i has q_i input strings and p_i output strings. Some of the input strings may be connected to some output strings, and strings may be permuted. We think of diagrams with q input strings and p output strings as representing a morphism $W^{\otimes q} \longrightarrow W^{\otimes p}$. An example with $q = 3 = p$ is



. (4.1)

Two diagrams are equivalent if and only if they define the same map for every algebraic structure of type $((p_i, q_i))$, cf. [Me1, Def. 4.1]. We define $\mathrm{Con}^{p,q}$ to be the vector space freely spanned by diagrams with q input strings and p output strings. In particular, $\mathrm{Con}^{0,0}$ is freely spanned by closed diagrams, and it has the structure of a commutative $\mathbb{K}$-algebra, where the multiplication is given by juxtaposition of diagrams. We write $\mathbb{K}[X]_{\mathrm{aug}} \equiv \mathbb{K}[X]_{\mathrm{aug}}((p_i, q_i))$ for the algebra $\mathrm{Con}^{0,0}$.

In [Me2] the universal category $\mathcal{C}_{\mathrm{univ}}$ generated by a dualisable object W with structure tensors of type $((p_i, q_i))$ was constructed. The objects in $\mathcal{C}_{\mathrm{univ}}$ are formal direct sums of objects of the form $W^{a,b} := W^{\otimes a} \otimes (W^*)^{\otimes b}$, and the Hom spaces are defined using the spaces $\mathrm{Con}^{p,q}$. The category $\mathcal{C}_{\mathrm{univ}}$ satisfies the following universal property:

Proposition 4.1. [Me2, Prop. 3.1] Let $\mathcal{D}$ be a $\mathbb{K}$-linear additive rigid symmetric monoidal category.[5] Evaluation at W gives a bijection between isomorphism classes of symmetric monoidal $\mathbb{K}$-linear functors $\mathcal{C}_{\mathrm{univ}} \longrightarrow \mathcal{D}$ and isomorphism classes of algebraic structures of type $((p_i, q_i))$ in $\mathcal{D}$.

We would like to consider here only algebraic structures that satisfy the axioms of a closed Λ_r-Frobenius algebra. All the axioms we consider can be thought of as the vanishing of certain elements in $\mathrm{Con}^{p,q}$ (for some $p, q \in \mathbb{N}$). A *theory* is a collection of axioms $\mathcal{T} \subseteq \bigsqcup_{p,q} \mathrm{Con}^{p,q}$. An algebraic structure of type $((p_i, q_i))$ is said to be a *model* of $\mathcal{T}$ if every element of $\mathcal{T}$ is zero when considered as a map between tensor powers of W. We consider here the theory $\mathcal{T}$ of closed Λ_r-Frobenius algebras. In [Me2, Sect. 3.1] a quotient $\mathcal{C}_{\mathrm{univ}}^{\mathcal{T}}$ of $\mathcal{C}_{\mathrm{univ}}$ was constructed which has the following universal property:

Proposition 4.2. [Me2, Prop. 3.5] Let $\mathcal{D}$ be a $\mathbb{K}$-linear additive rigid symmetric monoidal category. Evaluation at W gives a bijection between isomorphism classes of symmetric monoidal $\mathbb{K}$-linear functors $\mathcal{C}_{\mathrm{univ}}^{\mathcal{T}} \longrightarrow \mathcal{D}$ and isomorphism classes of closed Λ_r-Frobenius algebras in $\mathcal{D}$.

In particular, since $\mathcal{C}_{\mathrm{univ}}^{\mathcal{T}}$ is a quotient of $\mathcal{C}_{\mathrm{univ}}$, the endomorphism ring of $\mathbf{1}$ in $\mathcal{C}_{\mathrm{univ}}^{\mathcal{T}}$ is a quotient of $\mathbb{K}[X]_{\mathrm{aug}}$, which we denote by $\mathbb{K}[X]_{\mathrm{aug}}/I_{\mathcal{T}} = \mathbb{K}[X]_{\mathrm{aug}}^{\mathcal{T}}$.

Define now a symmetric monoidal category $\mathcal{E}_0$ in the following way: the objects of $\mathcal{E}_0$ are the same as the objects of Bord_2^r, and for $X, Y \in \mathcal{E}_0$ we have

$$\mathrm{Hom}_{\mathcal{E}_0}(X, Y) = \mathbb{K}\,\mathrm{Hom}_{\mathrm{Bord}_2^r}(X, Y)\,, \tag{4.2}$$

[5] In [Me2] the assumption of additivity is not explicit.

the free $\mathbb{K}$-vector space generated by the Hom set in Bord_2^r. Composition is defined in the obvious way. The symmetric monoidal structure on Bord_2^r induces a symmetric monoidal structure on $\mathcal{E}_0$. Take now $\mathcal{E}$ to be the idempotent completion of the additive closure of $\mathcal{E}_0$. Hence objects in $\mathcal{E}$ are direct summands of objects of the form $\bigoplus_{i=1}^n X_i$, where $X_i \in \mathcal{E}_0$, and

$$\mathrm{Hom}_{\mathcal{E}}\Big(\bigoplus_i X_i, \bigoplus_j Y_j\Big) = \bigoplus_{i,j} \mathrm{Hom}_{\mathcal{E}_0}(X_i, Y_j)\,. \tag{4.3}$$

Hom spaces between direct summands are defined in the obvious way. Write similarly $(\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}})_{+1}$ for the idempotent completion of $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}$.

Proposition 4.3. There is a natural equivalence of symmetric monoidal $\mathbb{K}$-linear categories between the category $(\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}})_{+1}$ and $\mathcal{E}$.

Proof. It is straightforward to see that if $\mathcal{D}$ is a $\mathbb{K}$-linear additive idempotent complete rigid symmetric monoidal category, then the category of symmetric monoidal functors $\mathrm{Bord}_2^r \longrightarrow \mathcal{D}$ is equivalent to the category of symmetric monoidal $\mathbb{K}$-linear functors $\mathcal{E} \longrightarrow \mathcal{D}$. By Theorem 2.9 we know that functors $\mathrm{Bord}_2^r \longrightarrow \mathcal{D}$ are equivalent to closed Λ_r-Frobenius algebras in $\mathcal{D}$. Proposition 4.2 above shows now that we have an equivalence of categories between functors from $\mathcal{E}$ to $\mathcal{D}$ and functors from $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}$ to $\mathcal{D}$. Since $\mathcal{D}$ is idempotent complete, these categories are also equivalent to the category of functors from $(\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}})_{+1}$ to $\mathcal{D}$. By taking $\mathcal{D} = \mathcal{E}$ and $\mathcal{D} = (\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}})_{+1}$ we get the desired equivalences. $\square$

It follows from the construction of $\mathcal{E}$ that the endomorphism ring of the unit object is the polynomial algebra on the set of all diffeomorphism classes of closed connected surfaces with an r-spin structure. By the equivalence above this algebra is isomorphic to $\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}$. Hence we have the identifications

$$\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} = \mathbb{K}[\{\beta_d\}_{d|r}, \alpha_1, \alpha_2, \dots] \qquad \text{if } r \text{ is odd,} \tag{4.4}$$

$$\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} = \mathbb{K}[\{\beta_d\}_{d|r}, \alpha_1^+, \alpha_1^-, \alpha_2^+, \alpha_2^-, \dots] \qquad \text{if } r \text{ is even} \tag{4.5}$$

for $r > 2$ (and we have to add a variable α_{-1} for the sphere invariant in case $r \in \{1, 2\}$).

Every closed Λ_r-Frobenius algebra in $\mathcal{D}$ gives rise to a functor $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}} \longrightarrow \mathcal{D}$, and hence in particular to a ring homomorphism $\mathrm{End}_{\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}}(\mathbf{1}) \longrightarrow \mathrm{End}_{\mathcal{D}}(\mathbf{1})$, that is, a map $\chi\colon \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathrm{End}_{\mathcal{D}}(\mathbf{1})$. If $\mathrm{End}_{\mathcal{D}}(\mathbf{1}) = \mathbb{K}$ we get $\chi\colon \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$. We call χ the *character of invariants* of the closed Λ_r-Frobenius algebra. We will think of such a χ as a function that assigns a numerical value to every diffeomorphism class of closed r-spin surfaces. Since $\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}$ is the polynomial algebra on the isomorphism classes of closed connected surfaces with an r-spin structure, such a character gives the scalar invariants that the TQFT corresponding to the closed Λ_r-Frobenius algebra associates to such surfaces.

4.2. **The categories $\mathcal{C}_\chi$.** Assume now that χ is a character as above, i.e. a ring homomorphism $\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$. To any two objects $X, Y \in \mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}$ we associate the pairing

$$\begin{aligned} \langle -,-\rangle\colon \mathrm{Hom}_{\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}}(X,Y) \otimes \mathrm{Hom}_{\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}}(Y,X) &\longrightarrow \mathrm{End}_{\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}}(\mathbf{1}) = \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \\ T_1 \otimes T_2 &\longmapsto \mathrm{tr}(T_1 \circ T_2)\,, \end{aligned} \tag{4.6}$$

and we define

$$\langle -,-\rangle_\chi := \chi \circ \langle -,-\rangle\,. \tag{4.7}$$

Definition 4.4. A morphism $f\colon X \longrightarrow Y$ in $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}$ is called *χ-negligible* if it is in the radical of the pairing $\langle -,-\rangle_\chi$. The subspace of χ-negligible morphisms is denoted $\mathcal{N}_\chi^{X,Y} \subseteq \mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}(X,Y)$.

The χ-negligible morphisms form a tensor ideal that we denote $\mathcal{N}_\chi$, see [Me2, Prop. 4.2].

Definition 4.5. The category $\mathcal{C}_\chi$ is the Karoubian envelope of the quotient $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}/\mathcal{N}_\chi$.

Proposition 4.6. The category $\mathcal{C}_\chi$ is tensor generated by a rigid object W that comes with the structure of a closed Λ_r-Frobenius algebra, and $\mathrm{End}_{\mathcal{C}_\chi}(\mathbf{1}) \cong \mathbb{K}$. If Σ is a closed surface with an r-spin structure, then the invariant that W assigns to Σ is exactly $\chi(\Sigma)$.

Proof. The category $\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}$ is already tensor generated by a rigid object W that is a closed Λ_r-Frobenius algebra. This property descents directly to the quotient $\mathcal{C}_\chi$. The collection of χ-negligible morphisms in $\mathrm{End}_{\mathcal{C}^{\mathcal{T}}_{\mathrm{univ}}}(\mathbf{1}) = \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}$ is exactly $\mathrm{Ker}(\chi)$ so we get that $\mathrm{End}_{\mathcal{C}_\chi}(\mathbf{1}) = \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}/\,\mathrm{Ker}(\chi) \cong \mathbb{K}$. Under the last isomorphism, the invariant that is associated to Σ is exactly the image of $\Sigma \in \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}$ inside $\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}}/\,\mathrm{Ker}(\chi)$, which is $\chi(\Sigma)$. □

This answers the question raised in Section 1. For every set of invariants we have constructed here a category with a closed Λ_r-Frobenius algebra that assigns exactly these values. However, the categories $\mathcal{C}_\chi$ are nicely behaved only for a specific collection of characters:

Definition 4.7. A category $\mathcal{C}$ is called $\mathbb{K}$-*good* if it is rigid, abelian, enriched over finite-dimensional $\mathbb{K}$-vector spaces, and $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong \mathbb{K}$.

Definition 4.8. A character $\chi\colon \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$ is called *good* if the following conditions are satisfied:

(1) All the Hom spaces in $\mathcal{C}_\chi$ are finite-dimensional.
(2) Every nilpotent endomorphism in $\mathcal{C}_\chi$ has trace zero.

The following is an adaptation of a result in [Me2] to closed Λ_r-Frobenius algebras.

Theorem 4.9. [Me2, Thm. 1.1] Let $\chi\colon \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$ be a character. The following conditions are equivalent:

(1) The category $\mathcal{C}_\chi$ is a semisimple $\mathbb{K}$-good category.
(2) The character χ is the character of invariants of some closed Λ_r-Frobenius algebra in some $\mathbb{K}$-good category.
(3) The character χ is a good character.

The above theorem already gives us a necessary condition for χ to be good:

Proposition 4.10. Let $\chi\colon \mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$ be a good character. If r is odd, then

$$\sum_{n\geqslant 0} \chi(\alpha_n) X^n \tag{4.8}$$

is a rational function that is contained in the space $R := \mathrm{span}\{\frac{1}{1-\lambda x}\}_{\lambda\in\mathbb{K}}$. If r is even, then both

$$\sum_{n\geqslant 0} \chi(\alpha_n^+) X^n \quad \text{and} \quad \sum_{n\geqslant 0} \chi(\alpha_n^-) X^n \tag{4.9}$$

are contained in R.

Proof. For the case r is odd and for the series $\sum_{n\geqslant 0}\chi(\alpha_n^+)X^n$ when r is even this follows from [Me2, Cor. 2.3], by considering the handle morphism, defined in Equation (2.29). In the last series we argue as follows: write $T_1 = (h^+_{x+2})^{\mathcal{Z}} \circ (h^+_{x+4})^{\mathcal{Z}} \circ \cdots \circ (h^+_{x-2})^{\mathcal{Z}} \circ (h^+_x)^{\mathcal{Z}}$ and $T_2 = (h^+_{x+2})^{\mathcal{Z}} \circ (h^+_{x+4})^{\mathcal{Z}} \circ \cdots \circ (h^+_{x-2})^{\mathcal{Z}} \circ (h^-_x)^{\mathcal{Z}}$. Then the series is actually

$$\sum_{n\geqslant 0} \mathrm{tr}(T_2^{n-1}T_1)X^n.$$

Lemma 2.10 implies that $T_1T_2 = T_2T_1$ and $T_1^2 = T_2^2$. We can then decompose the relevant space into eigenspaces for T_1 and T_2, and the result follows, again by [Me2, Cor. 2.3]. $\square$

4.3. **Example: the constant character and Deligne's category $\mathbf{Rep}(S_t)$.** We first consider the case $r = 1$, and let $t \in \mathbb{K}$ be any number. Consider the character χ_t that sends all the closed surfaces to t. In [Me2, Sect. 8] it is shown that χ_t is a good character, and that the resulting category $\mathcal{C}_{\chi_t}$ is the category $\mathrm{Rep}(S_t)$ that was introduced by Deligne in [De] as the interpolation of the representation categories of symmetric groups. By pulling back along $\mathrm{Bord}_2^r \longrightarrow \mathrm{Bord}_2^1$ we see that the constant character $\chi_t(\Sigma) = t$ is a good character also for arbitrary r. Intuitively, this algebra can be thought of as the commutative algebra $\mathbb{K}^t$ of all functions from the set with t elements to the ground field, with pointwise addition and multiplication. Of course, this makes actual sense only when $t \in \mathbb{N}$. The mere existence of the last algebra enables us to prove the following:

Lemma 4.11. The set of good characters $\mathbb{K}[X]^{\mathcal{T}}_{\mathrm{aug}} \longrightarrow \mathbb{K}$ forms a $\mathbb{K}$-algebra.

Proof. Assume that χ_1 and χ_2 are two good characters. Then there are categories $\mathcal{D}_1$ and $\mathcal{D}_2$ and closed Λ_r-Frobenius algebras W_1 and W_2 that afford these characters. Both algebras can be regarded as algebras inside the Deligne product $\mathcal{D}_1 \boxtimes \mathcal{D}_2$. As such, we can take their sums and products. By Lemma 3.9 we see that pointwise addition and pointwise multiplication of good characters is again a good character, as the category $\mathcal{D}_1 \boxtimes \mathcal{D}_2$ is again a good category. Since we also have an algebra with constant character values, this also shows that the set of good characters is closed under multiplication by a scalar $t \in \mathbb{K}$, and the set of good characters is indeed a $\mathbb{K}$-algebra under pointwise addition and multiplication of characters. $\square$

Remark 4.12. The above proof is similar to the proof of [Me2, Thm. 1.4]. A new proof is required here because the way the tensor product is defined is different from the setting of [Me2].

We are now ready to prove the following:

Theorem 4.13. If r is odd, then the character χ is good if and only if

$$\sum_{n\in\mathbb{Z}_{\geqslant 1}} \chi(\alpha_n)X^n \in \operatorname{span}\left\{\frac{1}{1-\lambda X}\right\}_{\lambda\in\mathbb{K}}. \tag{4.10}$$

If r is even, then the character χ is good if and only if only if

$$\sum_{n\in\mathbb{Z}_{\geqslant 1}} \chi(\alpha_n^+)X^n , \sum_{n\in\mathbb{Z}_{\geqslant 1}} \chi(\alpha_n^-)X^n \in \operatorname{span}\left\{\frac{1}{1-\lambda X}\right\}_{\lambda\in\mathbb{K}}. \tag{4.11}$$

Proof. We have already seen that this condition is necessary. To show that it is sufficient, we use the fact that the set of good characters form a $\mathbb{K}$-algebra. Consider the case where r is odd. By Lemma 3.7 and Lemma 3.2 the characters for the algebras E_μ, and $B^{(d)}$, with $d|r$, span the set of all characters that satisfies the condition of the theorem. We are done in this case. If r is even, we use the characters of the algebras A, $A\underline{\otimes}E_\mu$, and $C^{(d)}$ for $d|r$. By Equation (3.3), Lemma 3.7, and Lemma 3.5, they span the set of all characters that satisfy the above condition, and therefore we are done in this case as well. $\square$

Remark 4.14. All the algebras that we constructed in this paper can be realised inside a symmetric monoidal category of the form

$$\mathrm{SVect}_{\mathbb{K}} \boxtimes \bigboxtimes_i \mathrm{Rep}(S_{t_i}) \tag{4.12}$$

for some $t_i \in \mathbb{K}$. However, if χ is a good character, it is possible that the category $\mathcal{C}_\chi$ will not be of this form.

4.4. **Connection to the work of Khovanov, Ostrik, and Kononov.** In case $r = 1$ the question of possible values of the invariants for 2-dimensional TQFTs was studied in [KOK]. One gets the same numerical criterion [KOK, Thm. 3.7], where the question of abelian realisation of a given sequence was studied. The results in [KOK] also consider the case of a field of positive characteristic. The category $\mathcal{C}_\chi$ that we construct here is the same as the category denoted $\underline{\mathrm{DCob}}_\alpha$ in [KOK]. The category Cob_α appears in [Me2] as the category $\widetilde{\mathcal{C}}_\chi$.

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